# Granular Instrumental Variables

## From Micro Shocks to Macro Outcomes

Macro Lab Group Presentation

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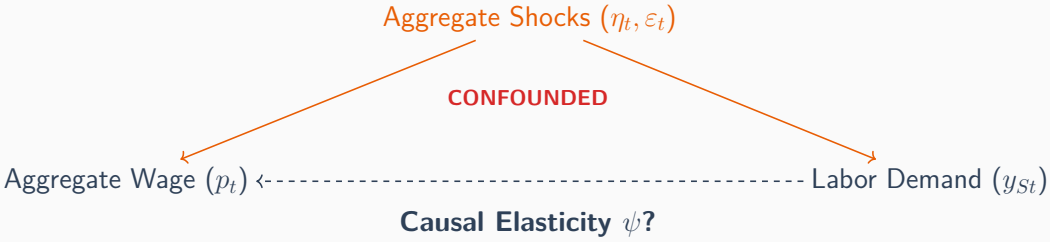
# The Law of Large Numbers fails in modern economies

In a standard economy, micro-shocks cancel out at a rate of $1/\sqrt{N}$.
In a **granular** economy, a few firms are so large that their

idiosyncratic luck drives aggregate volatility.

# The Endogeneity Trap prevents OLS identification



# Roadmap: From intuition to GMM estimation

1. **Act II, Sec 1:** The Simple Model — Difference-in-averages.
2. **Act II, Sec 2:** The Full Model — Heterogeneous factor loadings.
3. **Act II, Sec 3:** Implementation — Python construction and GMM.
4. **Act II, Sec 4:** Application — Superstar firms and wage spillovers.

# Section 1: The Simple Model

Difference-in-averages as an instrument

# Suppose Apple has a breakthrough, but tech is stagnant

Apple's idiosyncratic success ($u_{it}$) creates a labor demand shock. This shock is **uncorrelated** with macro-shocks, making it a

potential instrument for aggregate prices.

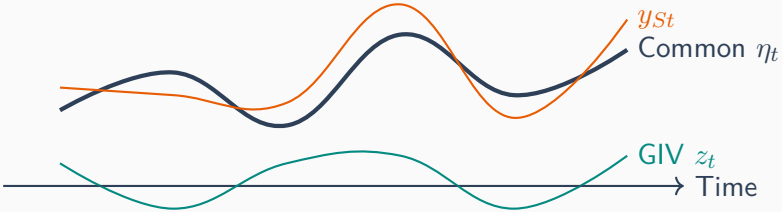
# Isolating the granular shock via size-weighting

The GIV ($z_t$) is the size-weighted average minus the equal-weighted average.

$$z_t = \underbrace{\sum S_i y_{it}}_{\text{Size-Weighted}} - \underbrace{\frac{1}{N}\sum y_{it}}_{\text{Equal-Weighted}}$$

If $y_{it} = \phi p_t + \eta_t + u_{it}$, then $z_t = u_{St} - u_{Et}$.

# Subtracting the average "kills" the common factor



# Section 2: The Full Model

Handling heterogeneous factor loadings

# Superstar firms respond differently to macro trends

The simple GIV ($y_{St} - y_{Et}$) assumes every firm has the same loading $\lambda_i = 1$. If large firms are more sensitive to the Fed, the simple GIV is **"leaky"**—it correlates with $\eta_t$.

# The Full Factor Model for granular firms

We now allow for heterogeneous loadings $\lambda_i$:

$$y_{it} = \phi p_t + \lambda_i \eta_t + u_{it} \tag{full}$$

The simple GIV becomes:

$$z_t = (\lambda_S - \lambda_E)\eta_t + (u_{St} - u_{Et})$$

If $\lambda_S \neq \lambda_E$, the instrument is **invalid** ($\mathbb{E}[z_t \eta_t] \neq 0$).

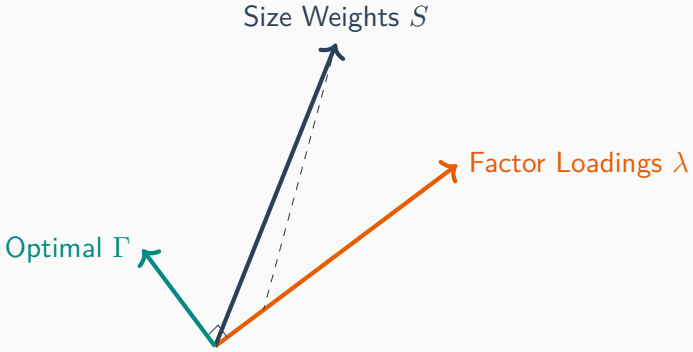
# The Solution: Orthogonal Optimal Weights $\Gamma$

We need weights $\Gamma$ such that the factor components **cancel exactly**.

$$\sum_{i=1}^{N} \Gamma_i \lambda_i = 0 \quad \text{(Orthogonality Condition)}$$

This ensures $z_t = \sum \Gamma_i u_{it}$ is purely idiosyncratic.

# Geometry of GIV: Projecting away from the macro trend




$\Gamma$ is the component of $S$ orthogonal to $\lambda$.

# Section 3: Implementation

Python construction and GMM estimation

# Constructing the GIV in Python

```python
def construct_giv(df, size_col, outcome_col):
    # 1. Weights
    df['S'] = df[size_col] / df.groupby('year')[size_col].
        transform('sum')
    df['E'] = 1.0 / df.groupby('year')['id'].transform('count')

    # 2. Simple GIV
    y_S = (df[outcome_col] * df['S']).groupby('year').sum()
    y_E = (df[outcome_col] * df['E']).groupby('year').sum()
    z = y_S - y_E

    return z
```

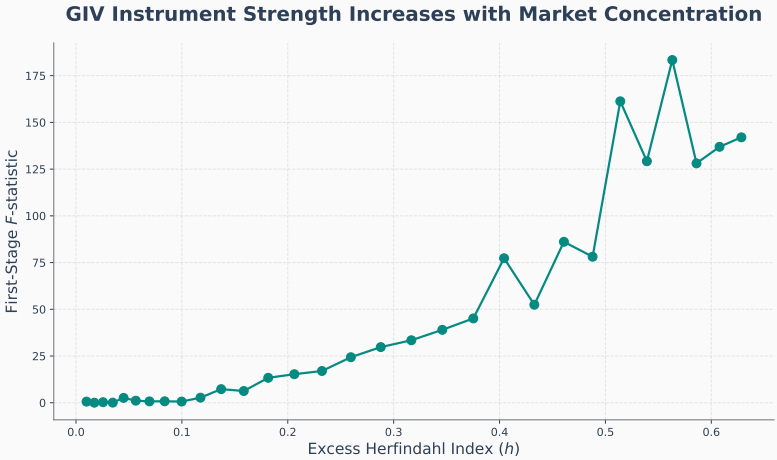
# GMM identifies the model when $\lambda$ is unknown

In practice, we estimate $\phi$ and $\lambda_i$ simultaneously via GMM. The moment conditions:

$$\mathbb{E}\left[(y_{it} - \phi p_t - \lambda_i \eta_t) \cdot z_t\right] = 0$$

We use the "Granular Residual" of the largest firms as our source of variation.

# Strength increases with concentration and micro-volatility



# Section 4: Labor Market Application

Superstar firms and wage spillovers

# Superstar firms drive local wage dynamics

When Amazon enters a city, its idiosyncratic demand shock ($u_{Amazon,t}$) shifts the aggregate labor demand. GIV identifies the

**General Equilibrium** response of local wages to this granular shift.

# Devil's Advocate: Are shocks truly idiosyncratic?

A skeptic would say: "Amazon's shock is just a hidden tech-sector trend." **The GIV Defense:**

- Include industry-year fixed effects (Common Factors).
- Use Narrative Checks (e.g., specific policy wins or product failures).
- Overidentification tests across size bins.

# GIV provides a bridge from micro data to macro identification

We no longer need to search for natural experiments in the aggregate data. The experiments are happening every day at the firm level—we just need to aggregate them correctly.

**Large firms' idiosyncratic shocks provide a valid, granular instrument to identify aggregate elasticities.**